

Introduction to Electronics Engineering

MODULE 2 — PART 2: QUESTION BANK

Module 2 (10 hrs): Bipolar Junction Transistors (BJT) & Junction Field Effect Transistors (JFET)

For First-Year Students — with complete answers and diagrams
Course: Introduction to Electronics Engineering (IEE) · Course Code: EC 101

Question Paper Structure

Section	Marks per Question	Scope	Questions
Section A	2 marks	Compulsory — from every module	Q1 – Q19
Section B	5 marks	Only from Modules 1 and 2	Q1 – Q12
Section C	7 marks	Compulsory — from every module	Q1 – Q7

- Every section covers **all the topics** of Module 2 — no topic is left out.
- No question is repeated between sections.
- Answer 2-mark questions in 2–4 lines, 5-mark questions with a diagram in about half a page, and 7-mark questions with full explanation and diagrams.
- Numerical examples included where the topic demands them.

SECTION A — 2-Mark Questions

Q1. What is a transistor? Why is the BJT called a bipolar device?

Answer: A transistor is a three-terminal semiconductor device used as an amplifier and as a switch. In a BJT, current conduction takes place due to both types of charge carriers — electrons and holes — hence it is called a **bipolar** junction transistor.

Q2. What are the three terminals of a BJT? State the function of each.

Answer: **Emitter (E):** heavily doped — supplies majority carriers to the base. **Base (B):** lightly doped and very thin — passes most carriers on to the collector. **Collector (C):** moderately doped — collects the majority carriers coming from the emitter.

Q3. What are the two types of BJT? How do they differ in construction?

Answer: The NPN transistor (emitter and collector are n-type; base is p-type) and the PNP transistor (emitter and collector are p-type; base is n-type). In an NPN, conduction is mainly by electrons; in a PNP, mainly by holes.

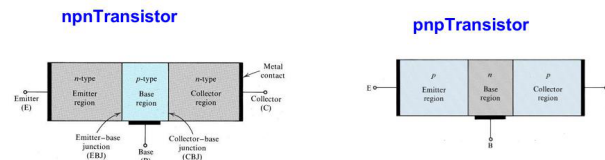


Figure 1: NPN and PNP transistor structures.

Q4. What are the three regions of operation of a BJT?

Answer: **Cut-off:** EB junction reverse biased, CB junction reverse biased — transistor OFF. **Active:** EB junction forward biased, CB junction reverse biased — used for amplification. **Saturation:** EB junction forward biased, CB junction forward biased — transistor fully ON.

Q5. Why is the BJT called a current-controlled device?

Answer: In a BJT the output current (I_C) is controlled by the input current (I_B) through the relation $I_C = \beta \times I_B$. The controlling quantity is a current, not a voltage.

Q6. Write the relation between the transistor currents.

Answer: By Kirchhoff's current law at the transistor: $I_E = I_B + I_C$, where I_E is the emitter current, I_B the base current and I_C the collector current.

$$I_C = \text{Collector current}$$

$$I_B = \text{Base current}$$

$$\beta \gg 1 \text{ (typically } 20 - 300\text{)}$$

Figure 2: The three transistor currents.

Q7. Define the current amplification factor alpha. What is its typical range?

Answer: Alpha is the current gain of the common base configuration: $\alpha = I_C / I_E$. Since $I_C < I_E$, alpha is always less than 1 (typically 0.95 to 0.99).

Q8. Define beta. How is it related to alpha?

Answer: Beta is the current gain of the common emitter configuration: $\beta = I_C/I_B$ (typically 20–500).
 Relation: $\beta = \alpha/(1 - \alpha)$ and $\alpha = \beta/(1 + \beta)$.

Q9. What are ICBO and ICEO? How are they related?

Answer: I_{CBO} is the collector-base leakage current with the emitter open; I_{CEO} is the collector-emitter leakage current with the base open. They are related by $I_{CEO} = (\beta + 1) \times I_{CBO}$, so I_{CEO} is much larger.

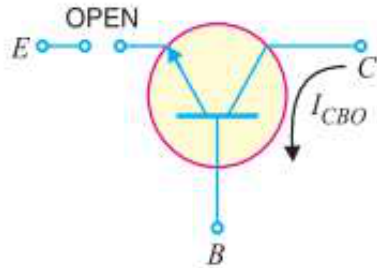


Figure 3: Leakage current I_{CBO} (emitter open).

Q10. What is the Early effect (base-width modulation)?

Answer: When the reverse bias voltage V_{CB} of the collector-base junction increases, the depletion region widens and the effective base width reduces. This slight reduction in base width is the Early effect; it reduces recombination and shifts the input characteristic curves.

Q11. Why is the common emitter configuration the most widely used?

Answer: Because it provides both high current gain (β) and high voltage gain, with moderate input and output impedances — ideal for general-purpose amplification. Its output voltage is 180° phase-inverted relative to the input.

Q12. What is the common collector (CC) configuration used for? What is its other name?

Answer: It is used for impedance matching and as a buffer stage, because of its high input impedance, low output impedance and approximately unity voltage gain. It is also called the **emitter follower**.

Q13. What is transistor biasing? Why is it needed?

Answer: Biasing is the process of setting the DC operating voltage and current (the Q-point) of a transistor to the correct level, so that the EB junction is forward biased and the CB junction reverse biased — keeping the transistor in the active region for faithful amplification.

Q14. What is a DC load line? Write its equation and end-points.

Answer: The DC load line is the graphical line of all possible I_C – V_{CE} combinations for a given V_{CC} and R_C : $V_{CE} = V_{CC} - I_C R_C$. End-points: $I_{C(max)} = V_{CC}/R_C$ (at $V_{CE} = 0$) and $V_{CE(max)} = V_{CC}$ (at $I_C = 0$).

Q15. What is the Q-point? Where should it be located for faithful amplification?

Answer: The Q-point (quiescent point) is the DC operating point — the intersection of the DC load line with the output characteristic for the given base current. Ideally it lies at the **midpoint of the load line**, so the output can swing symmetrically without distortion or clipping.

Q16. Why is the JFET called a unipolar and voltage-controlled device?

Answer: In a JFET, current conduction is due to only one type of carrier (electrons in an n-channel, holes in a p-channel) — hence unipolar. The drain current is controlled by the gate-source **voltage** V_{GS} — hence voltage-controlled.

Q17. Name the three terminals of a JFET and state their functions.

Answer: **Source (S):** the terminal through which the majority carriers enter the channel. **Drain (D):** the terminal through which they leave. **Gate (G):** controls the width of the channel (and hence the drain current) through a reverse-biased junction.

Q18. Define pinch-off voltage and I_{DSS} .

Answer: **Pinch-off voltage (V_P)** is the drain voltage at which the channel is completely pinched off, beyond which I_D becomes nearly constant. I_{DSS} is the drain-source saturation current — the maximum drain current, obtained when $V_{GS} = 0$ V.

Q19. Write Shockley's equation for the JFET.

Answer: $I_D = I_{DSS} \times (1 - V_{GS}/V_P)^2$, where I_D is the drain current, V_{GS} the gate-source voltage and V_P the pinch-off voltage.

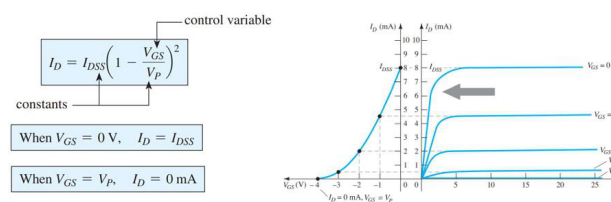


Figure 4: Shockley's JFET equation.

SECTION B — 5-Mark Questions**Q1. Explain the construction and symbol of NPN and PNP transistors with diagrams.**

Answer:

NPN: a thin, lightly doped p-type base is sandwiched between two n-type layers (emitter and collector). The emitter is heavily doped and the collector moderately doped. The circuit symbol has the arrow on the emitter pointing **outward** (arrow = NPN, "Not Pointing iN").

PNP: a thin, lightly doped n-type base is sandwiched between two p-type layers. The emitter arrow points **inward** toward the base. Operation is complementary: holes are the majority carriers and the bias polarities are reversed.

In both types, the base is made very thin and lightly doped so that almost all carriers injected by the emitter pass through to the collector.

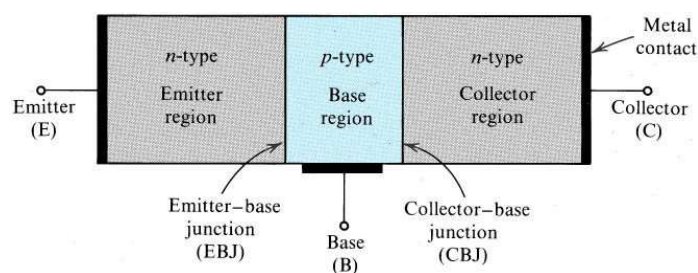


Figure 5: NPN transistor cross-section.

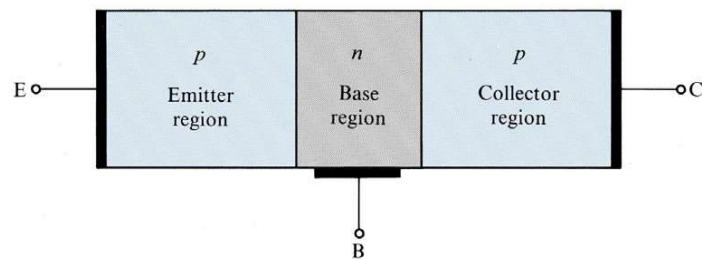


Figure 6: PNP transistor cross-section.

Q2. Explain the working of an NPN transistor (with the bias circuit diagram).

Answer:

- Bias: the emitter-base junction is forward biased and the collector-base junction is reverse biased (active region).
- The forward-biased EB junction makes electrons in the emitter flow toward the base — this is the emitter current I_E .
- In the thin, lightly doped base, very few electrons recombine with holes — this small recombination is the base current I_B .
- The remaining electrons cross the base and are swept through the collector by the reverse-biased CB junction — the collector current I_C .
- Since I_C is much larger than I_B , the transistor provides current amplification: $I_E = I_B + I_C$.

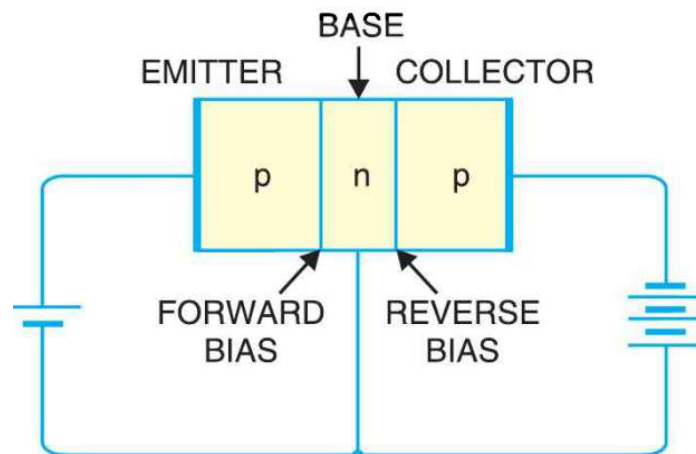


Figure 7: NPN transistor in the active region.

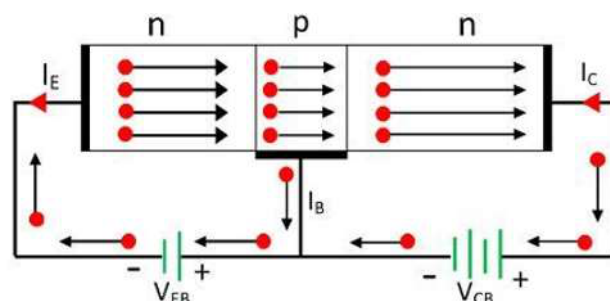


Figure 8: Current flow in the NPN transistor.

Q3. Explain the working of a PNP transistor.**Answer:**

- Bias: the EB junction is forward biased and the CB junction reverse biased — but all supply polarities are the reverse of the NPN case.
- The forward-biased EB junction makes holes in the p-type emitter flow toward the base — this is the emitter current I_E .
- In the thin, lightly doped n-type base, very few holes recombine with electrons — this small recombination is the base current I_B .
- The remaining holes cross the depletion region and move through the collector to the negative terminal of the external DC source — the collector current I_C .
- The current relations are identical to the NPN: $I_E = I_B + I_C$.

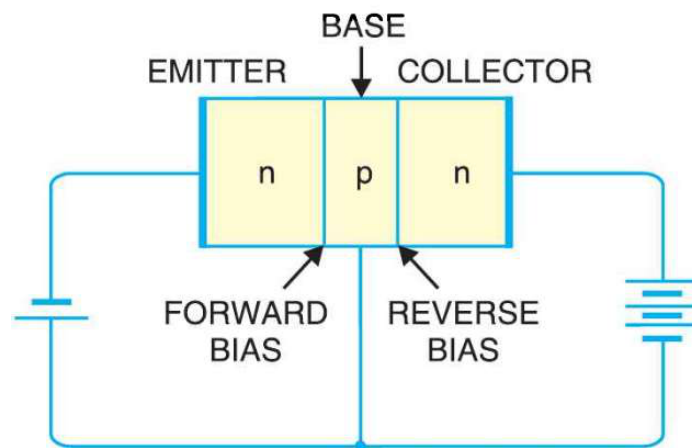


Figure 9: PNP transistor in the active region.

Q4. Draw the input and output characteristics of the common base (CB) configuration and explain them.**Answer:**

Input characteristics: I_E versus V_{BE} at constant V_{CB} . The curve looks like a forward-biased diode: a small increase in V_{BE} causes a rapid rise in I_E . A larger V_{CB} shifts the curve slightly left (Early effect).

Output characteristics: I_C versus V_{CB} at constant I_E . In the active region I_C is nearly constant and approximately equal to I_E ($\alpha < 1$). At low (negative) V_{CB} is the saturation region; for $I_E = 0$ only the leakage current I_{CBO} flows (cut-off).

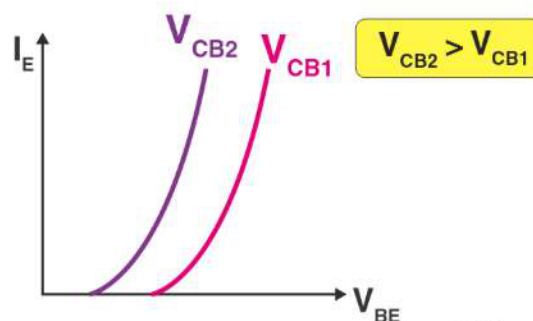


Figure 10: CB input characteristics.

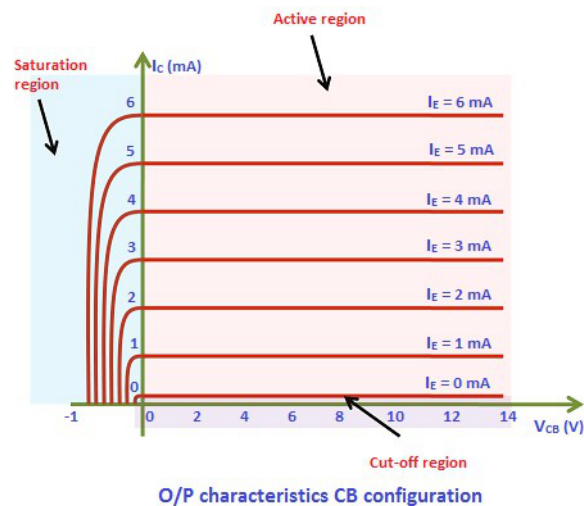


Figure 11: CB output characteristics.

Q5. Draw the input and output characteristics of the common emitter (CE) configuration and explain the three regions.

Answer:

Input characteristics: I_B versus V_{BE} at constant V_{CE} — like a forward-biased diode curve.

Output characteristics: I_C versus V_{CE} for different I_B values:

- **Active region:** BE junction forward biased, CB junction reverse biased; I_C is nearly constant and equal to $\beta \times I_B$.
- **Saturation region:** at very low V_{CE} both junctions are forward biased and I_C rises steeply.
- **Cut-off region:** below $I_B = 0$, only the leakage current I_{CEO} flows.
- Current relation: $I_C = \beta I_B + I_{CEO}$.

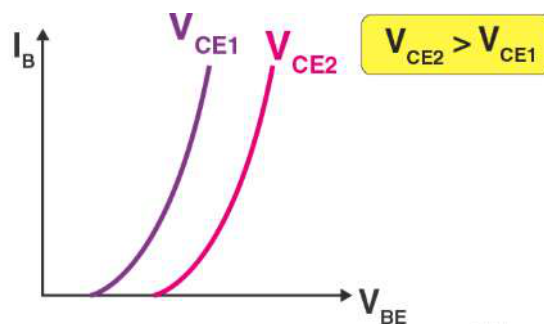


Figure 12: CE input characteristics.

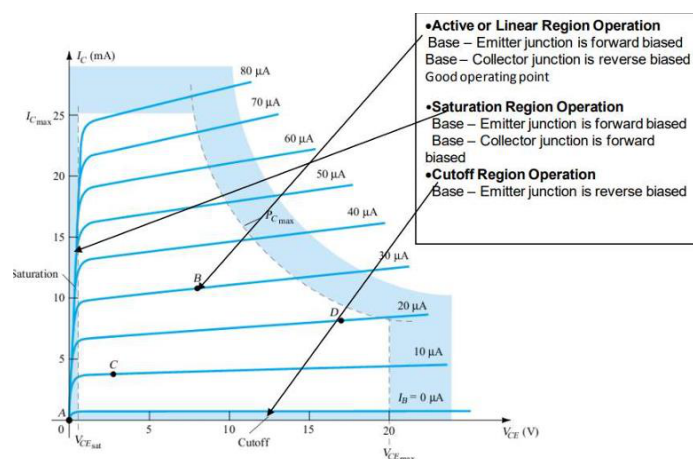


Figure 13: CE output characteristics with the three regions.

Q6. Explain the common collector (CC) configuration and its characteristics.**Answer:**

- The collector is common to the input and output circuits; input is applied to the base and the output is taken from the emitter.
- **Input characteristics:** I_B versus V_{CB} at constant V_{CE} .
- **Output characteristics:** I_E versus V_{CE} at constant I_B — similar in shape to the CE output curves, but with I_E as the output current.
- Properties: high input impedance, low output impedance, approximately unity voltage gain and **no** phase inversion.
- Applications: impedance matching and buffer (isolation) stages.

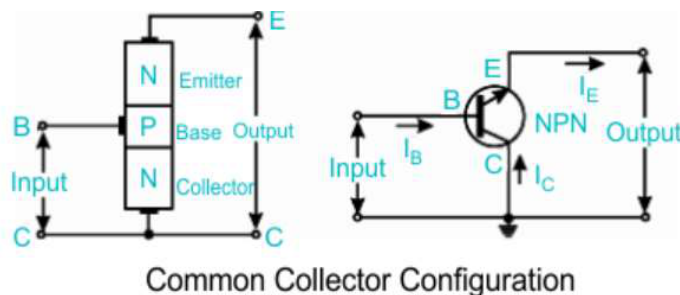


Figure 14: Common collector circuit.

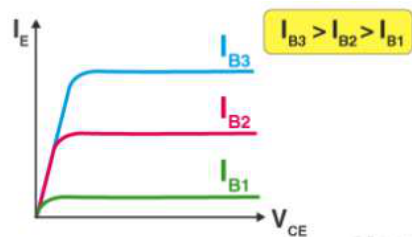


Figure 15: CC output characteristics.

Q7. Derive the relations between alpha, beta, I_{CBO} and I_{CEO} .**Answer:**

Start with the CB collector-current equation and KCL:

$$I_C = \alpha I_E + I_{CBO} \quad \text{and} \quad I_E = I_B + I_C$$

Substituting I_E :

$$\begin{aligned} I_C &= \alpha (I_B + I_C) + I_{CBO} \\ I_C (1 - \alpha) &= \alpha I_B + I_{CBO} \\ I_C &= [\alpha / (1 - \alpha)] I_B + [1 / (1 - \alpha)] I_{CBO} \end{aligned}$$

Comparing with the CE equation $I_C = \beta I_B + I_{CEO}$:

$$\beta = \alpha / (1 - \alpha) \quad \text{and} \quad I_{CEO} = (1 / (1 - \alpha)) I_{CBO} = (\beta + 1) I_{CBO}$$

Also: $\alpha = \beta / (1 + \beta)$ and $1 / (1 - \alpha) = \beta + 1$.

$$I_E = I_C + I_B$$

$$\alpha = \frac{I_C}{I_E}, \quad \beta = \frac{I_C}{I_B}$$

$$\Rightarrow \beta = \frac{I_C}{I_E - I_C} = \frac{\alpha I_E}{I_E(1 - \alpha)}$$

$$\boxed{\beta = \frac{\alpha}{1 - \alpha}}$$

$$\boxed{\alpha = \frac{\beta}{1 + \beta}}$$

$$\left[\because \frac{1}{1 - \alpha} = \beta + 1 \right]$$

Figure 16: The alpha-beta relations.

Q8. Explain the fixed bias method with circuit diagram and DC analysis.

Answer:

In the fixed bias circuit, a single resistor R_B connects V_{CC} to the base, and R_C connects V_{CC} to the collector; the emitter is grounded.

Base-emitter loop (KVL): $V_{CC} - I_B R_B - V_{BE} = 0$, so:

$$I_B = (V_{CC} - V_{BE}) / R_B$$

Collector current: $I_C = \beta \times I_B$

Collector-emitter loop (KVL):

$$V_{CE} = V_{CC} - I_C R_C$$

The circuit is simple, but the Q-point is unstable because I_C depends directly on β (which varies with temperature and from device to device).

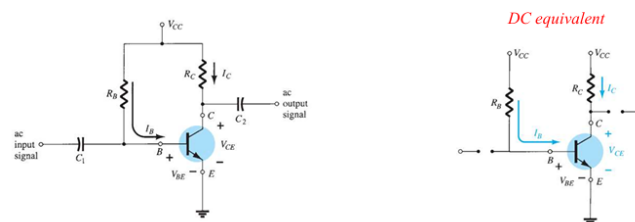


Figure 17: The fixed bias circuit.

Q9. Explain the emitter bias method with circuit diagram and equations.

Answer:

The emitter bias circuit adds an emitter resistor R_E between the emitter and ground. The voltage drop across R_E increases with I_E , which reduces V_{BE} and thereby opposes any change in collector current — the circuit self-stabilises.

Base-emitter loop (KVL), with $I_E = (\beta + 1)I_B$:

$$I_B = (V_{CC} - V_{BE}) / [R_B + (\beta + 1) R_E]$$

Collector-emitter loop (KVL), with $I_E \approx I_C$:

$$V_{CE} = V_{CC} - I_C (R_C + R_E)$$

This arrangement gives much better stability than the fixed bias.

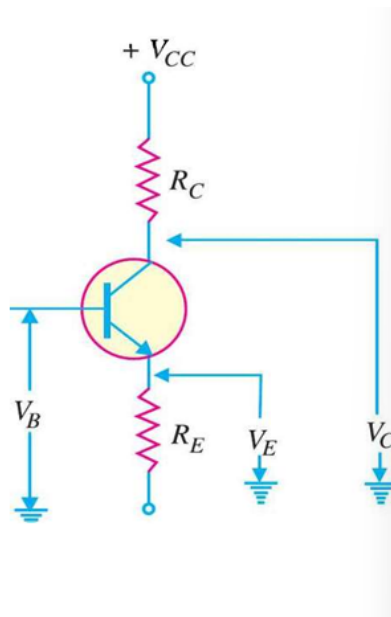


Figure 18: The emitter bias circuit.

Q10. Explain the voltage divider bias with circuit diagram and Thevenin equivalent.

Answer:

Resistors R_1 and R_2 form a voltage divider across V_{CC} to fix the base voltage, so the Q-point depends very little on beta.

The network to the left of the base is replaced by its Thevenin equivalent:

$$R_{Th} = R_1 \parallel R_2 \quad (\text{with the supply replaced by a short circuit})$$

$$V_{Th} = V_{CC} \times R_2 / (R_1 + R_2) \quad \text{in series with } R_{Th}$$

Because the divider holds the base voltage constant, and R_E stabilises the emitter current, this is the most stable and the most widely used bias arrangement.

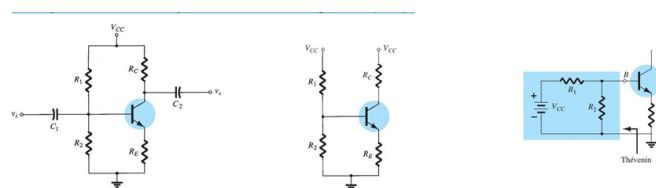


Figure 19: Voltage divider bias circuit and its Thevenin equivalent.

Q11. Explain the construction and working of the n-channel JFET.

Answer:

Construction: an n-type silicon bar (the channel) has two p-type regions diffused on opposite sides; these are joined internally and brought out as the gate. Ohmic contacts at the ends of the bar form the source and the drain.

Working:

- With V_{DD} applied between drain and source (gate at 0 V), electrons flow through the channel from source to drain.

- For small V_{DS} the channel behaves like a resistor: I_D rises linearly with V_{DS} .
- As V_{DS} increases, the depletion layers at the gate junctions widen, narrowing the channel until it is pinched off; beyond this, I_D saturates (I_{DSS}).
- Applying a reverse gate-source voltage V_{GS} widens the depletion layers further, reducing the drain current; a positive-going V_{GS} toward zero increases it.

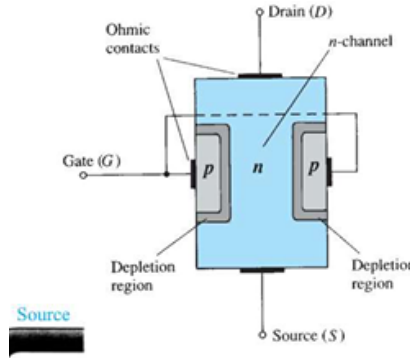


Figure 20: Construction of the n-channel JFET.

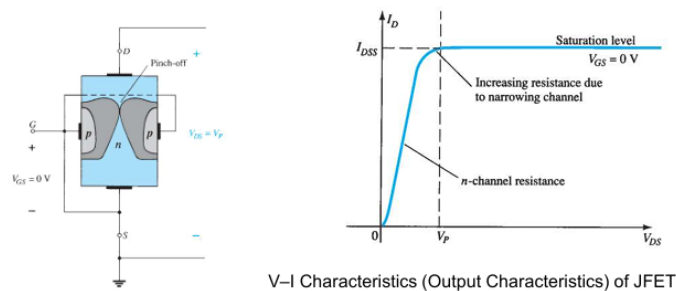


Figure 21: Drain characteristics of the JFET.

Q12. Explain the transfer characteristics of the JFET and derive the key points from Shockley's equation.

Answer:

The transfer characteristic is the graph of I_D versus the control voltage V_{GS} , governed by Shockley's equation:

$$I_D = I_{DSS} (1 - V_{GS}/V_P)^2$$

- At $V_{GS} = 0$ V: $I_D = I_{DSS} (1 - 0)^2 = I_{DSS}$ — the maximum drain current.
- At $V_{GS} = 0.3 V_P$: $I_D = I_{DSS} (1 - 0.3)^2 = I_{DSS}/2$.
- At $V_{GS} = 0.5 V_P$: $I_D = I_{DSS} (1 - 0.5)^2 = I_{DSS}/4$.
- At $V_{GS} = V_P$: $I_D = 0$ mA — the channel is fully pinched off.
- The curve is a parabola, symmetric about the V_{GS} axis in shape, defining the operating range 0 to V_P .

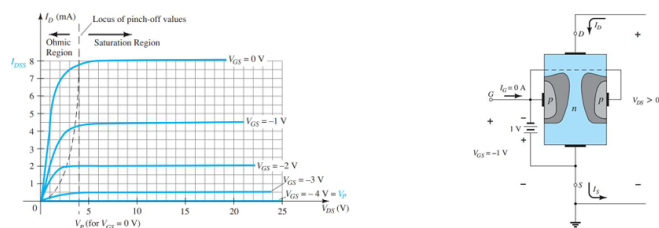


Figure 22: JFET transfer characteristics.

SECTION C — 7-Mark Questions

Q1. Describe the structure of the BJT and explain the working of NPN and PNP transistors in detail (with diagrams).

Answer:

Structure: a BJT has three doped regions forming two PN junctions: the emitter (heavily doped), the base (thin and lightly doped) and the collector (moderately doped, largest area). The two junctions share the base region.

NPN working: with the EB junction forward biased and the CB junction reverse biased, the emitter injects electrons into the base. Only a small fraction recombine in the thin base (this is the small base current I_B); the rest diffuse across and are collected (I_C). The emitter current is the sum: $I_E = I_B + I_C$.

PNP working: identical, except holes are the carriers: the emitter injects holes into the n-type base, few recombine with electrons (I_B), and the rest are collected (I_C). All bias polarities are reversed compared to the NPN.

Because $I_C \gg I_B$, a small base current controls a much larger collector current — the basis of transistor amplification.

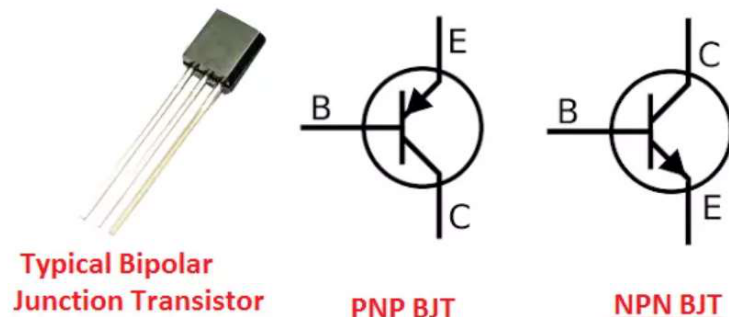


Figure 23: BJT terminals and symbols.

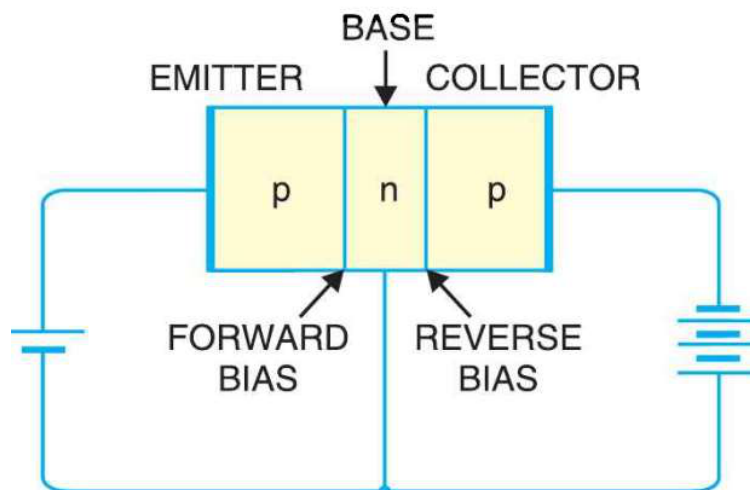


Figure 24: NPN transistor biasing and operation.

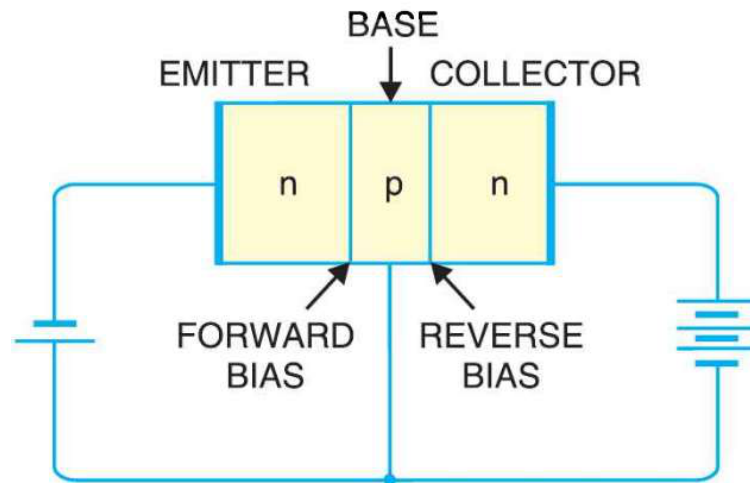


Figure 25: PNP transistor biasing and operation.

Q2. Explain the CB, CE and CC configurations of the BJT with diagrams and compare them.

Answer:

CB: base common; input at the emitter, output at the collector. Voltage gain high, current gain $\alpha < 1$, low input impedance, no phase inversion; used at high frequencies.

CE: emitter common; input at the base, output at the collector. High current gain (beta) and high voltage gain; output inverted 180°; the most used configuration for amplification.

CC (emitter follower): collector common; input at the base, output at the emitter. Voltage gain about 1, high input and low output impedance; used for impedance matching and buffering.

Parameter	CB	CE	CC
Common terminal	Base	Emitter	Collector
Current gain	Less than 1 (α)	High (β)	High ($\beta + 1$)
Voltage gain	High	High	About 1
Input impedance	Low	Medium	High
Output impedance	High	Medium	Low
Phase shift	0°	180°	0°
Application	High-frequency	General amplification	Buffer / matching

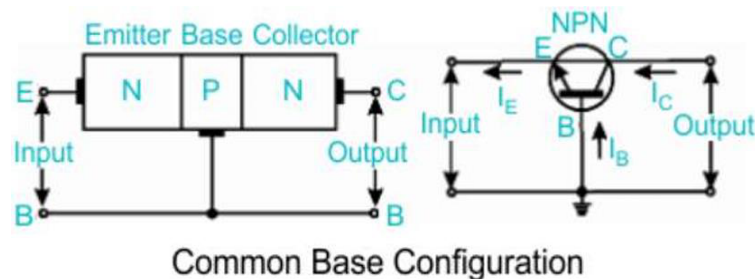


Figure 26: Common base configuration.

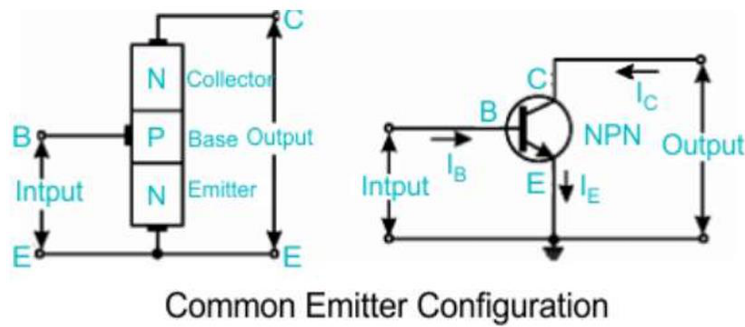


Figure 27: Common emitter configuration.

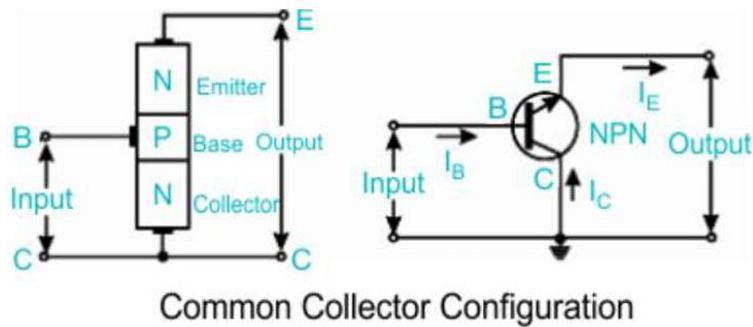


Figure 28: Common collector configuration.

Q3. Draw and explain in detail the input and output characteristics of the CE configuration. Mark and explain all three regions of operation.

Answer:

Input characteristics (I_B vs V_{BE} at constant V_{CE}): the curve resembles a forward-biased diode. Beyond the knee (~ 0.7 V for Si), I_B rises rapidly. Increasing V_{CE} shifts the curves slightly right due to the Early effect.

Output characteristics (I_C vs V_{CE} for different I_B):

- **Cut-off region:** below the $I_B = 0$ curve; both junctions reverse biased; only the leakage current I_{CEO} flows.
- **Active region:** the flat portion of the curves; BE junction forward biased, CB junction reverse biased; $I_C = \beta I_B$ and is nearly independent of V_{CE} — the transistor amplifies here.
- **Saturation region:** the steeply rising part at very low V_{CE} ; both junctions forward biased; further increase in I_B gives no increase in I_C .
- Current relation over the whole plane: $I_C = \beta I_B + I_{CEO}$.

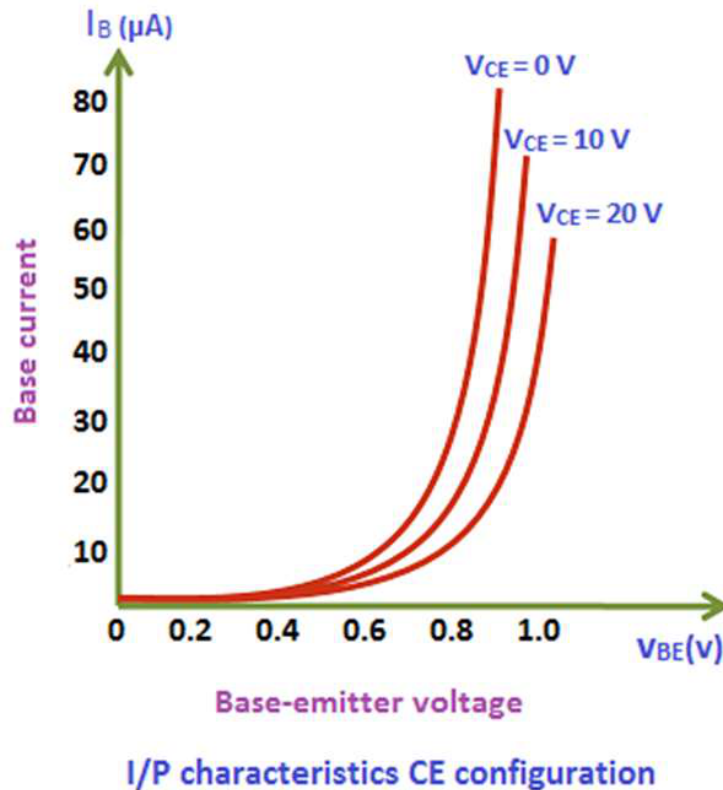


Figure 29: CE input characteristic curves.

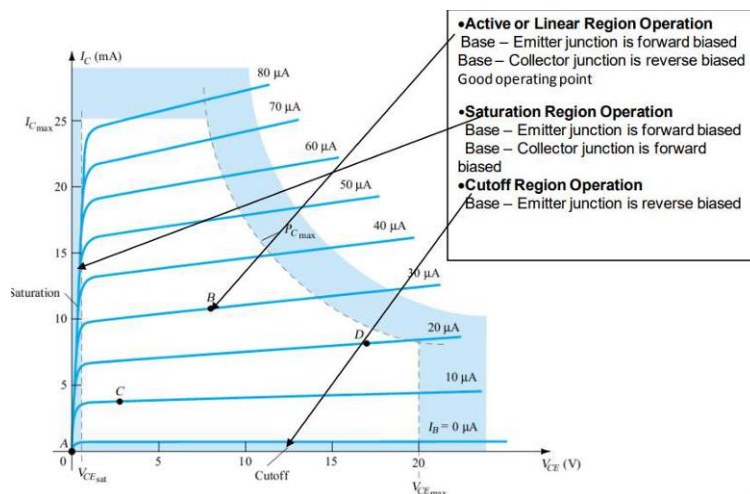


Figure 30: CE output characteristics with the three regions.

Q4. Explain the DC load-line analysis and Q-point of a BJT in detail. How do the biasing methods (fixed, emitter, voltage divider, collector feedback) compare?

Answer:

Load line: from KVL in the output loop, $V_{CE} = V_{CC} - I_C R_C$. This straight line joins the points $I_{C(max)} = V_{CC}/R_C$ ($V_{CE} = 0$) and $V_{CE(max)} = V_{CC}$ ($I_C = 0$) on the output characteristics.

Q-point: the intersection of the load line with the curve for the actual base current — the DC operating point. Ideally at the **midpoint** of the load line for maximum symmetrical output swing. If the Q-point is too near saturation, the negative peaks clip; if too near cut-off, the positive peaks clip.

Biasing methods:

- **Fixed bias:** one resistor R_B ; simple but unstable (Q-point shifts with beta).

- **Emitter bias:** adds R_E ; the emitter-voltage feedback opposes Q-point drift.
- **Voltage divider bias:** R_1 – R_2 divider fixes the base voltage; almost independent of beta — the best stability, most used.
- **Collector feedback bias:** R_B from collector to base; self-stabilising with a single supply.

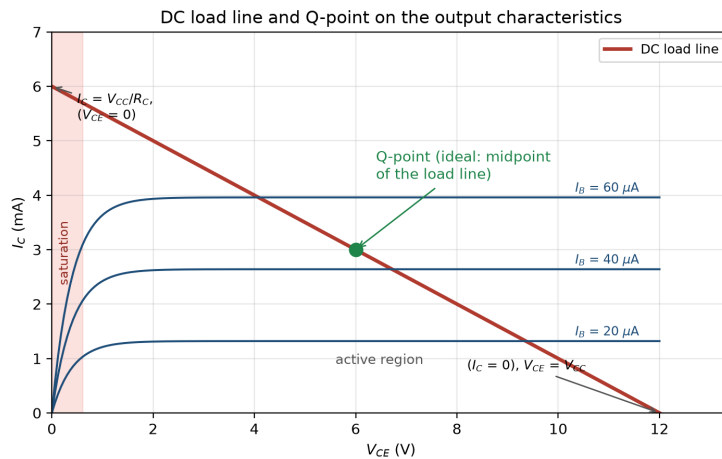


Figure 31: DC load line and Q-point.

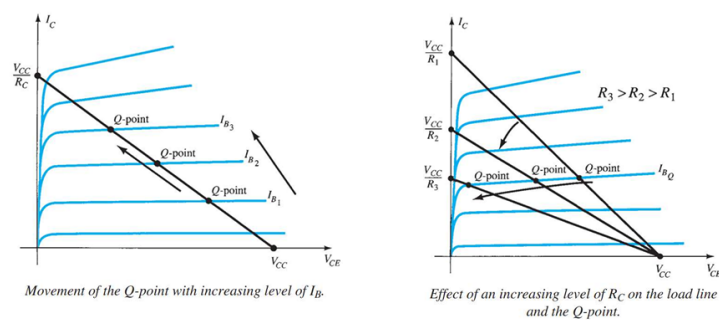
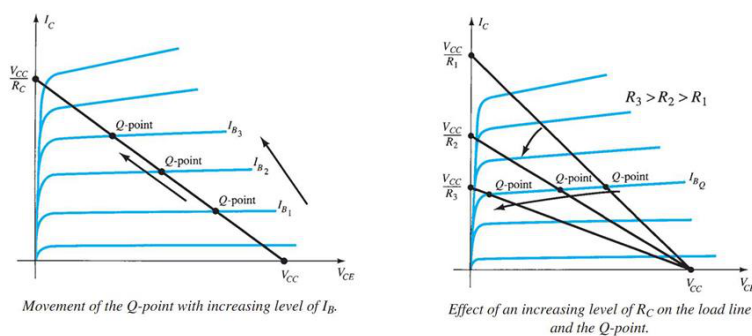


Figure 32: Movement of the Q-point with base current.

Figure 33: Effect of R_C and V_{CC} on the load line.

Q5. Explain the construction, working, pinch-off voltage and characteristics (drain and transfer) of the JFET in detail.

Answer:

Construction: n-channel bar with two p-type gate regions diffused on the sides; the gates are joined to one terminal. Source and drain are ohmic contacts at the bar ends.

Working: with $V_{GS} = 0$ and increasing V_{DS} , I_D first rises linearly (ohmic region), then the depletion layers constrict the channel until **pinch-off**, after which I_D saturates at a nearly constant value. A reverse V_{GS} reduces I_D further, up to $V_{GS} = V_P$ where $I_D = 0$.

Drain (output) characteristics: I_D vs V_{DS} for different V_{GS} ; curves for more negative V_{GS} sit lower, until pinch-off for all curves.

Transfer characteristics: I_D vs V_{GS} — a parabola from I_{DSS} (at $V_{GS} = 0$) to zero (at $V_{GS} = V_P$), defined by Shockley's equation $I_D = I_{DSS}(1 - V_{GS}/V_P)^2$.

Pinch-off voltage (V_P): the value of V_{DS} at which the channel is fully constricted and I_D becomes constant.

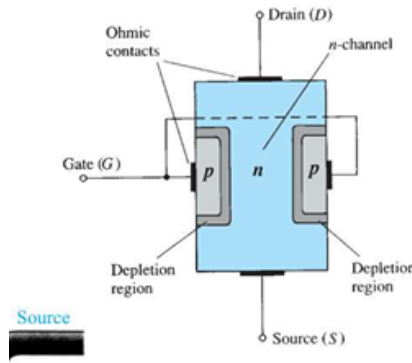
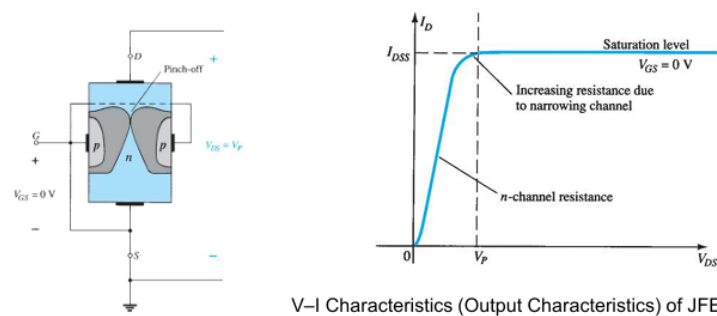


Figure 34: JFET construction.



V-I Characteristics (Output Characteristics) of JFET

Figure 35: Drain characteristics.

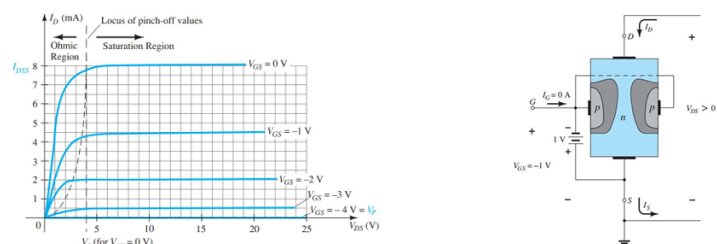


Figure 36: Transfer characteristics.

Q6. Compare the BJT and the JFET in detail (construction, control, characteristics, advantages, applications).

Answer:

Parameter	BJT	JFET
Type	Bipolar, current-controlled	Unipolar, voltage-controlled
Terminals	Base, Emitter, Collector	Gate, Source, Drain
Carriers	Electrons and holes	One type only
Input impedance	Low to medium	Very high
Noise	Higher	Lower
Gain	High current and voltage gain	Lower voltage gain
Key equation	$I_C = \beta I_B + I_{CEO}$	$I_D = I_{DSS}(1 - V_{GS}/V_P)^2$
Biasing of control junction	Forward biased (EB)	Reverse biased (gate-source)
Applications	Amplifiers, fast switching, logic	Low-noise amplifiers, analog switches

Because the JFET gate is reverse biased, it draws almost no gate current — giving it a very high input impedance, ideal for buffers and low-noise front ends. The BJT, in exchange, offers higher gain and faster switching for general amplification.



Figure 37: JFET circuit symbols.

Q7. Numerical: In a common base connection, $I_E = 1.2$ mA and $\alpha = 0.96$. Find I_C and I_B . Also find beta.

Answer:

Given: $I_E = 1.2$ mA, $\alpha = 0.96$.

Step 1 — collector current: $I_C = \alpha \times I_E = 0.96 \times 1.2 = 1.152$ mA

Step 2 — base current: $I_B = I_E - I_C = 1.2 - 1.152 = 0.048$ mA = 48 μ A

Step 3 — current gain: $\beta = \alpha / (1 - \alpha) = 0.96 / 0.04 = 24$

Check: $I_C = \beta \times I_B = 24 \times 48 \mu\text{A} = 1.152$ mA — consistent.

$$\alpha = \frac{I_C}{I_E}$$

I_C = Collector current

I_E = Emitter current

$\alpha < 1$ (typically 0.95 – 0.99)

$$I_E = I_C + I_B$$

Figure 38: Current relations used.